

A conversation on quantum information

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The game plan

- ▶ Start from the familiar abstraction: a **classical bit** and gates acting on it.
- ▶ Replace the bit by a vector: a **qubit**.
- ▶ Replace Boolean gates by **unitary matrices**.
- ▶ Put qubits together with the **tensor product**.
- ▶ A quantum resource: **entanglement**.
- ▶ Build a Bell pair, trace out a qubit, and compute a bit (!) of **entanglement entropy**.

Only prerequisite

Comfort with vectors, matrices, eigenvalues, and matrix multiplication.

Classical warm-up: a bit is an abstract state

A classical bit has two possible states:

$$0, \quad 1.$$

A gate is a rule that transforms those states. For example,

$$\text{NOT} : \quad 0 \leftrightarrow 1.$$

We can encode the two states as vectors,

$$0 \longleftrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad 1 \longleftrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

so NOT becomes

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Preview

Quantum computing keeps this linear-algebra viewpoint and then refuses to restrict the state vector to just the two basis vectors.

A minimal rulebook for quantum mechanics

1. **State:** an isolated system is represented by a normalized vector $|\psi\rangle$ in a complex inner-product space.
2. **Evolution:** a closed system changes by a unitary matrix U :

$$|\psi\rangle \mapsto U |\psi\rangle, \quad U^\dagger U = \mathbb{1}.$$

3. **Measurement:** amplitudes become probabilities; in an orthonormal basis $\{|i\rangle\}$,

$$p(i) = |\langle i|\psi\rangle|^2.$$

4. **Composition:** systems combine using a tensor product,

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B.$$

What is a qubit?

Choose two orthonormal basis states,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A qubit is any normalized vector in $\mathbb{C}^2$:

$$\boxed{|\psi\rangle = \alpha |0\rangle + \beta |1\rangle}, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Examples:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Important

α and β are *amplitudes*, not probabilities. The probabilities are their squared magnitudes after choosing a measurement basis.

One qubit: phase and the Bloch sphere

Write

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

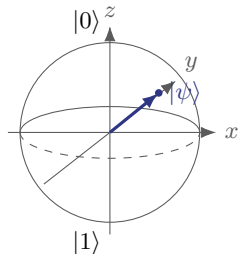
Multiplying the entire state by a **global phase** changes nothing observable:

$$|\psi\rangle \sim e^{i\gamma} |\psi\rangle.$$

$\Rightarrow$ states are really “rays.” One nice decomposition:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle.$$

the **Bloch sphere**.



Measurement: amplitudes become classical outcomes

Suppose

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle.$$

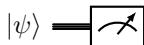
Measure in the computational basis $\{|0\rangle, |1\rangle\}$:

$$\Pr(0) = |\alpha|^2, \quad \Pr(1) = |\beta|^2.$$

After the measurement, the state is the observed basis state:

$$0 \Rightarrow |0\rangle, \quad 1 \Rightarrow |1\rangle.$$

Denote this as:



Basis dependent e.g., take $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, measure either in $\{|0\rangle, |1\rangle\}$, or $\{|+\rangle, |-\rangle\}$. Very different results!

Quantum gates are unitary matrices

A one-qubit gate is a 2×2 unitary matrix. The Pauli gates are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Their action on basis states:

$$X |0\rangle = |1\rangle, \quad X |1\rangle = |0\rangle,$$

$$Z |0\rangle = |0\rangle, \quad Z |1\rangle = -|1\rangle.$$

Classical analogy

X is literally NOT on the computational basis. Z has no ordinary one-bit Boolean analogue because it changes *phase* rather than the observed bit value.

Hadamard: switch between two useful bases

The Hadamard gate is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

It maps

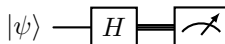
$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle,$$

and also

$$H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle.$$

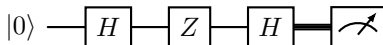
So measuring in the $\{|+\rangle, |-\rangle\}$ basis is equivalent to:

apply $H \longrightarrow$ measure in $\{|0\rangle, |1\rangle\}$.



A circuit is matrix multiplication read from left to right

Consider



The state evolves as

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{Z} |-\rangle \xrightarrow{H} |1\rangle.$$

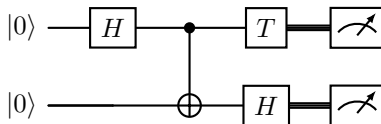
Algebraically,

$$|\psi_{\text{out}}\rangle = HZH |0\rangle = X |0\rangle = |1\rangle.$$

Order matters

The circuit is drawn in time order left-to-right, while the corresponding matrix product acts right-to-left on the state vector.

The circuit model



Read a circuit as a sequence of mathematical operations:

- ▶ wires carry quantum states;
- ▶ boxes are unitary matrices;
- ▶ multi-wire boxes can create **entanglement** (???);
- ▶ meters turn quantum amplitudes into classical outcomes.

The output of a circuit is the classical distribution obtained upon a chosen measurement.

A tale of two qubits

Two classical bits have four possible strings:

$$00, 01, 10, 11 \in \{0, 1\} \times \{0, 1\}$$

Two qubits live in

$$\mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4,$$

with computational basis

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

A general pure two-qubit state is

$$|\psi\rangle = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle,$$

where

$$\sum_{x \in \{00, 01, 10, 11\}} |\alpha_x|^2 = 1.$$

For n qubits, the state vector has 2^n complex amplitudes.

A tale of two qubits: tensor products

Alice and Bob hold a qubit each:

$$|\psi\rangle_A = \alpha |0\rangle + \beta |1\rangle, \quad |\phi\rangle_B = \gamma |0\rangle + \delta |1\rangle,$$

then

$$|\psi\rangle_A \otimes |\phi\rangle_B = \alpha\gamma |00\rangle + \alpha\delta |01\rangle + \beta\gamma |10\rangle + \beta\delta |11\rangle.$$

Example:

$$|+\rangle \otimes |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$

A state that can be written as $|\psi\rangle_A \otimes |\phi\rangle_B$ is a **product state**. What if I have a state $|\Psi\rangle \in \mathbb{C}^4$ that *cannot* be written as a product?

Quantum Entanglement

Consider the “Bell state”

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Can we find $a, b, c, d \in \mathbb{C}$ such that

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = \frac{|00\rangle + |11\rangle}{\sqrt{2}}?$$

Matching coefficients would require

$$ac \neq 0, \quad bd \neq 0, \quad ad = 0, \quad bc = 0,$$

which is impossible.

Definition

A pure bipartite state that is not a product state is **entangled**.

CNOT: the simplest two qubit gate

On computational-basis states, CNOT acts like a classical controlled XOR:

$$|x, y\rangle \longmapsto |x, y \oplus x\rangle.$$

Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 11\rangle$
$ 11\rangle$	$ 10\rangle$

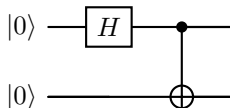
As a matrix,

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Because quantum mechanics is linear, this same rule automatically specifies what CNOT does to every superposition.

Making a Bell pair

Start from two qubits in $|00\rangle$:



After the Hadamard on the first qubit,

$$|00\rangle \mapsto \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}.$$

Then CNOT gives

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}} \mapsto \boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}}} = |\Phi^+\rangle.$$

A one-qubit superposition plus a two-qubit interaction created entanglement.

Measuring a Bell pair

For

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}},$$

computational-basis measurement $m = (m_0, m_1)$ gives

$$\Pr(00) = \frac{1}{2}, \quad \Pr(11) = \frac{1}{2}, \quad \Pr(01) = \Pr(10) = 0.$$

$m_0 = m_1$, fully correlated. However, individual statistics are fully random:

$$\Pr(m_0 = 0) = \Pr(m_0 = 1) = \frac{1}{2}$$

What is the mathematical formalism that one can use to talk only about the first qubit?

The density matrix formalism

For a pure state $|\psi\rangle$, define its density matrix

$$\rho = |\psi\rangle\langle\psi|.$$

If $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then

$$\rho = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}.$$

The diagonal entries look like probabilities; the off-diagonal entries store **coherence**. Unitary evolution becomes

$$\rho \longmapsto U\rho U^\dagger.$$

Density matrices are especially useful when we want to describe only *part* of a larger system.

Looking at only one half: the partial trace

Suppose ρ_{AB} describes qubits A and B . The state seen by someone with access only to A is

$$\rho_A = \text{Tr}_B(\rho_{AB}).$$

On basis operators,

$$\text{Tr}_B(|a\rangle\langle a'| \otimes |b\rangle\langle b'|) = |a\rangle\langle a'| \langle b'|b\rangle.$$

So terms with different B -labels vanish under the partial trace.

Interpretation

“Trace out B ” means: describe all predictions available locally on A , without pretending we know or control B .

Reduced state of a Bell pair

Start from

$$\rho_{AB} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle + |11\rangle)(\langle 00| + \langle 11|).$$

Expand:

$$\rho_{AB} = \frac{1}{2} \left(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11| \right).$$

Trace out qubit B :

$$\rho_A = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{\mathbb{1}}{2}.$$

A striking contrast

The joint two-qubit state is pure, yet either qubit by itself is “maximally mixed”.

Entropy: a number for uncertainty in a quantum state

For a density matrix ρ , the von Neumann entropy is

$$S(\rho) = -\text{Tr}(\rho \log_2 \rho).$$

If ρ has eigenvalues λ_i , then

$$S(\rho) = -\sum_i \lambda_i \log_2 \lambda_i.$$

This is exactly Shannon entropy applied to the eigenvalue distribution.
Two useful landmarks:

$$\rho = |\psi\rangle\langle\psi| \quad \Rightarrow \quad S(\rho) = 0,$$

$$\rho = \frac{\mathbb{1}}{2} \quad \Rightarrow \quad S(\rho) = 1 \text{ bit}.$$

Entanglement entropy of the Bell pair

For a bipartite *pure* state $|\psi\rangle_{AB}$, define the entanglement entropy across $A|B$ by

$$S_A = S(\rho_A), \quad \rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|).$$

For $|\Phi^+\rangle$, we found

$$\rho_A = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Its eigenvalues are $\frac{1}{2}, \frac{1}{2}$, so

$$S_A = -2 \left(\frac{1}{2} \log_2 \frac{1}{2} \right) = \boxed{1}.$$

One Bell pair contains one **e**bit of entanglement across the cut.

Compare: a product state has zero entanglement entropy

Take

$$|\psi\rangle_{AB} = |+\rangle_A \otimes |+\rangle_B.$$

Then

$$\rho_{AB} = |+\rangle\langle+|_A \otimes |+\rangle\langle+|_B.$$

Tracing out B gives

$$\rho_A = |+\rangle\langle+|,$$

a rank-one projector. Its eigenvalues are 1, 0, so

$$S(\rho_A) = 0.$$

For pure bipartite states

$$\text{product state} \iff S(\rho_A) = 0, \quad \text{entangled state} \iff S(\rho_A) > 0.$$

An immediate outcome: no-cloning

Could there be a unitary U that copies every unknown qubit?

$$U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle .$$

Apply this to copying the $|+\rangle$ state, we get,

$$U (|+\rangle |0\rangle) = |+\rangle |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle),$$

Now note that $U (|0\rangle |0\rangle) = |0\rangle |0\rangle$ and $U (|1\rangle |0\rangle) = |1\rangle |1\rangle$. By linearity of quantum mechanics,

$$U \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \neq U (|+\rangle |0\rangle)$$

No-cloning theorem

There is no physical operation that perfectly copies an arbitrary quantum state.

Going further...

Natural next conversations:

- ▶ Quantum computation: Clifford gates, magic, universal gate-sets, Solovay-Kitaev...
- ▶ Quantum algorithms: phase kickback, oracles, Grover search, and Fourier transforms, quantum advantages...
- ▶ Quantum many-body physics: matrix product states, tensor networks, classical simulation algorithms...
- ▶ Quantum information theory: entanglement for mixed states, distillation, LOCC...
- ▶ Noise and quantum channels: open systems, decoherence...
- ▶ Quantum error correction: route to scalable computation, fault tolerance...

Suggested references: Nielsen and Chuang, *Quantum Computation and Quantum Information* and Watrous, *Theory of quantum information*.